IMAGE PROCESSING TECHNIQUES – EXTENDED CLASS ACTIVITIES WORKSHEET

UG Level | Duration: 10 Minutes per Activity

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Activity 17

The "Blob" Hunt (FAST Detector Game)

- Concept: The FAST (Features from Accelerated Segment Test) algorithm defines a corner if a set of continuous pixels in a circle are all brighter or darker than the center pixel.
- Activity:
    1. Create a grid of cells, with some cells containing a "pixel" (a dot) of varying intensity (1-10).
    2. Participants (the "detector") look for a central pixel and check its 16 surrounding pixels in a circle.
    3. They mark the point as a corner if at least 9 (or another defined threshold) of the surrounding pixels are significantly brighter or darker than the center.

Concept: The FAST (Features from Accelerated Segment Test) algorithm identifies a corner by examining the 16 surrounding pixels arranged in a circle around a central pixel. A point is considered a corner when a sufficient number of continuous surrounding pixels are significantly brighter or darker than the center pixel.

Observation/Result: If at least 9 surrounding pixels satisfy the brightness condition compared with the central pixel, the central point can be marked as a corner. This activity demonstrates how FAST quickly identifies distinctive points in an image by comparing pixel intensities.